

Optimization Problems Practice

Instructions

Complete within 60 minutes. Show setup and key steps. Use exact values unless a decimal approximation is explicitly requested.

Basic Formulae

- Critical points occur where $f'(x) = 0$ or where $f'(x)$ is undefined.
- Rectangle: $A = lw$, perimeter $P = 2l + 2w$
- Box: $V = lwh$
- Cylinder: $V = \pi r^2 h$, surface area $S = 2\pi r^2 + 2\pi r h$
- Cone: $V = \frac{1}{3}\pi r^2 h$
- Sphere: $V = \frac{4}{3}\pi r^3$, surface area $S = 4\pi r^2$
- For similar triangles, corresponding side ratios are equal.

Problems

1. A rectangle is inscribed in the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

with sides parallel to the coordinate axes. Find the dimensions of the rectangle of maximum area.

2. A rectangle is inscribed under the parabola

$$y = 18 - 2x^2$$

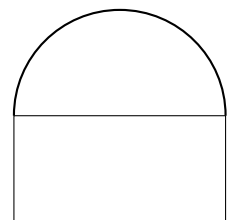
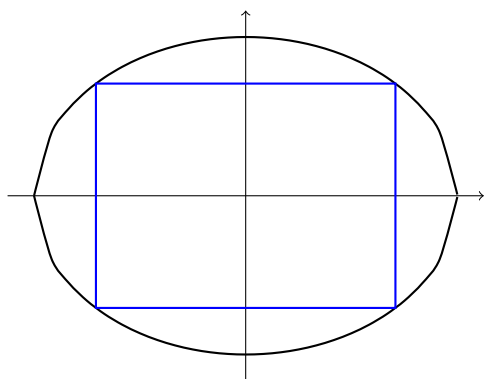
and above the x -axis, with sides parallel to the axes. Find the dimensions of the rectangle of maximum area.

3. A Norman window consists of a rectangle topped by a semicircle. If the total perimeter is 24 feet, find the dimensions that maximize the area.
4. A box with a square base and no top must have volume 768 cubic centimeters. Find the dimensions that minimize the amount of material used.
5. A closed cylinder must have volume 500π cubic centimeters. Find the radius and height that minimize the surface area.
6. A page is to contain 150 square inches of print. The side margins are 1 inch each and the top and bottom margins are 1.5 inches each. Find the dimensions of the page with minimum area.
7. A wire 30 units long is cut into two pieces. One piece is bent into an equilateral triangle and the other into a circle. How should the wire be cut so that the total enclosed area is minimized?
8. An open-top box is made by cutting equal squares of side length x from each corner of a 24 by 36 sheet of cardboard and folding up the sides. Find the value of x that maximizes the volume.
9. A right circular cone is inscribed in a sphere of radius R . Find the height of the cone of maximum volume.
10. A trough is formed from a 16-foot wide strip of metal by bending up two sides of equal width 4 feet, making an isosceles trapezoidal cross section. Find the angle that maximizes the cross-sectional area.
11. A ladder 20 feet long leans against a vertical wall. A rectangle is formed with one side on the ground, one side on the wall, and its upper-right corner touching the ladder. Find the dimensions of the rectangle of maximum area.
12. Find the point on the parabola

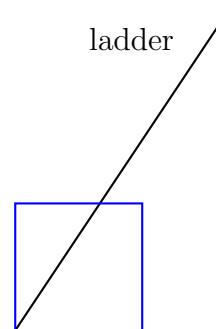
$$y = x^2$$

that is closest to the point $(0, 3)$. Use the distance formula.

Diagrams and Answers



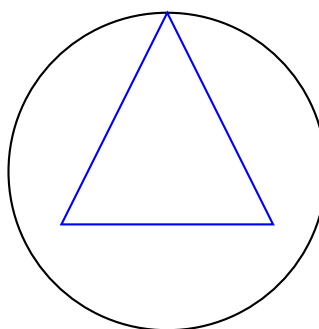
Norman window



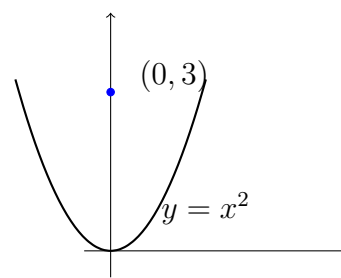
ladder



trough cross section



cone in sphere



Answers

1. $4\sqrt{2} \times 3\sqrt{2}$

2. width = $2\sqrt{3}$, height = 12

3. window width = $\frac{48}{4 + \pi}$, rectangular height = $\frac{24}{4 + \pi}$

4. base side = $8\sqrt[3]{3}$, height = $4\sqrt[3]{3}$

5. $r = 5\sqrt[3]{2}$, $h = 10\sqrt[3]{2}$

6. page dimensions = 12 by 18

7. triangle perimeter = $\frac{90\sqrt{3}}{\pi + 3\sqrt{3}}$, circle circumference = $30 - \frac{90\sqrt{3}}{\pi + 3\sqrt{3}}$

8. $x = 10 - 2\sqrt{7}$

9. $h = \frac{4R}{3}$

10. bend angle satisfies $\cos \theta = \frac{\sqrt{3}-1}{2}$, so $\theta = \arccos\left(\frac{\sqrt{3}-1}{2}\right)$

11. rectangle dimensions = $5\sqrt{2}$ by $5\sqrt{2}$

12. closest points are $\left(\pm\sqrt{\frac{5}{2}}, \frac{5}{2}\right)$